... the filter by using 'freqs( )' function

...01:pi

...n returns complex frequency response vector 'h' and frequency vector 'om' in radians/samples of the filter.

8. Calculate the magnitude of the frequency response in decibels (dB m=20*log10(abs(h))
9. Plot the magnitude response [magnitude in dB Vs normalized frequency (om/pi)]
10. Calculate the phase response using an = angle(h)
11. Plot the phase response [phase in radians Vs normalized frequency (om/pi)]
12. Give relevant names to x and y axes and give an appropriate title for the plot.
13. Plot all the responses in a single figure window.[Make use of subplot]

PROCEDURE

1. Open MATLAB
2. Open new M-file
3. Type the program
4. Save in current directory
5. Compile and Run the program
6. For the output see command window/Figure window

MATLAB PROGRAM

LP IIR Filter

```
clc;
clear all;
close all;
rp = input ('enter the passband ripple:');
rs = input ('enter the stopband ripple:');
fp = input ('enter the passband freq:');
```



```
fs = input ('enter the stopband freq:');
f = input ('sampling freq:');
wp = 2 * pi * (fp/f);
ws = 2 * pi * (fs/f);
[n, wc] = buttord (rp, rs, wp, ws, 's');
[b, a] = butter (n, wc, 'low', 's');
w = 0:0.01:pi;
[h, om] = freqs (b, a, w);
m = 20 * log (abs(h));
subplot (2,1,1);
plot (om/pi, m);
title ('magnitude response of input low pass
                filter');
xlabel ('normalized freq ');
ylabel ('gain in db');
an = angle(h);
subplot (2,1,2);
plot (om/pi, an);
title ('phase response of low pass filter');
xlabel('normalized frequency');
ylabel('phase in degree');
```

PF

$\alpha_p = 3dB$

$\alpha_s = 20dB$

$f_p = 1000\,Hz$ $\quad f_s = 2000\,Hz$

$f = 5000.$

$\Omega_p = 2\pi f_p T$

$= 2 * \pi * \frac{1000}{5000} = \frac{2}{5}\pi = 0.4\pi$

$\Omega_s = 2\pi f_s T = 0.8\pi$

$$N \geq \frac{\log\left(\sqrt{\frac{10^{0.1\alpha_s}-1}{10^{0.1\alpha_p}-1}}\right)}{\log\left(\frac{\Omega_s}{\Omega_p}\right)} \Rightarrow N \geq \frac{\log\left(\sqrt{\frac{10^{0.1\times 20}-1}{10^{0.1\times 3}-1}}\right)}{\log\left(\frac{0.8\pi}{0.4\pi}\right)}$$

$N \geq 3.3181$

$\boxed{N = 4}$

Step2:- Round off $N = 4$

Step3:- factor or poly of $N = 4$

$(s^2 + 0.765375 + 1)(s^2 + 1.84775 + 1)$

Step4:- $\Omega_c = \frac{\Omega_p}{(10^{0.1\alpha_p}-1)^{\frac{1}{8}}} = 1.25738$

Step5:- sub $s = \frac{s}{\Omega_c} = \frac{s}{1.25738}$

$$H(s) = \frac{(1.25738)^4}{(s^2 + 0.962365 + 1.58101)(s^2 + 2.921235 + 1.58101)}$$

$s = j\omega$

$$H(j\omega) = \frac{(1.25738)^4}{((j\omega)^2 + 0.962365(j\omega) + 1.58101)((j\omega)^2 + 2.92123(j\omega) + 1.58101(j\omega))}.$$

```
fs = input('ente
f = input('sar
wp = 2 * pi * (
ws = 2 * pi * (
[n, wc] = butt
[b, a] = butto
w = 0:0.01:p
[h, om] = freqs
m = 20 * log(
subplot(2,1,1
plot(om/pi, m
title('magni
xlabel('nam
ylabel('gai
an = angle(
subplot(2,1
plot(om/p
title('phas
xlabel('nam
ylabel('ph
```

enter the passband ripple : 0.2

enter the stopband ripple : 50

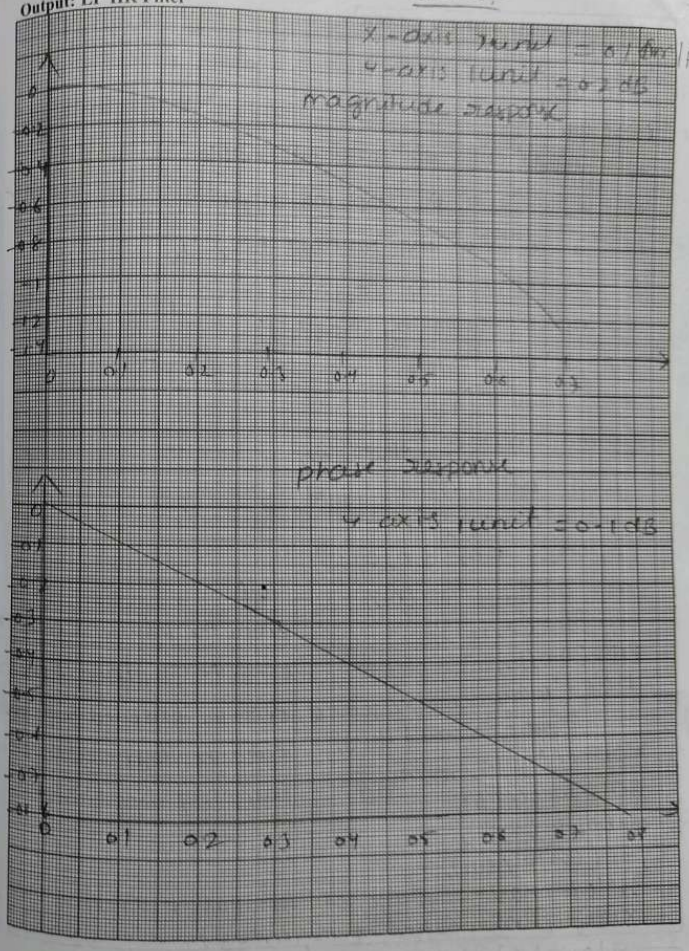
enter the passband freq : 5000

enter the stopband freq : 2000

Sampling freq : 5000.

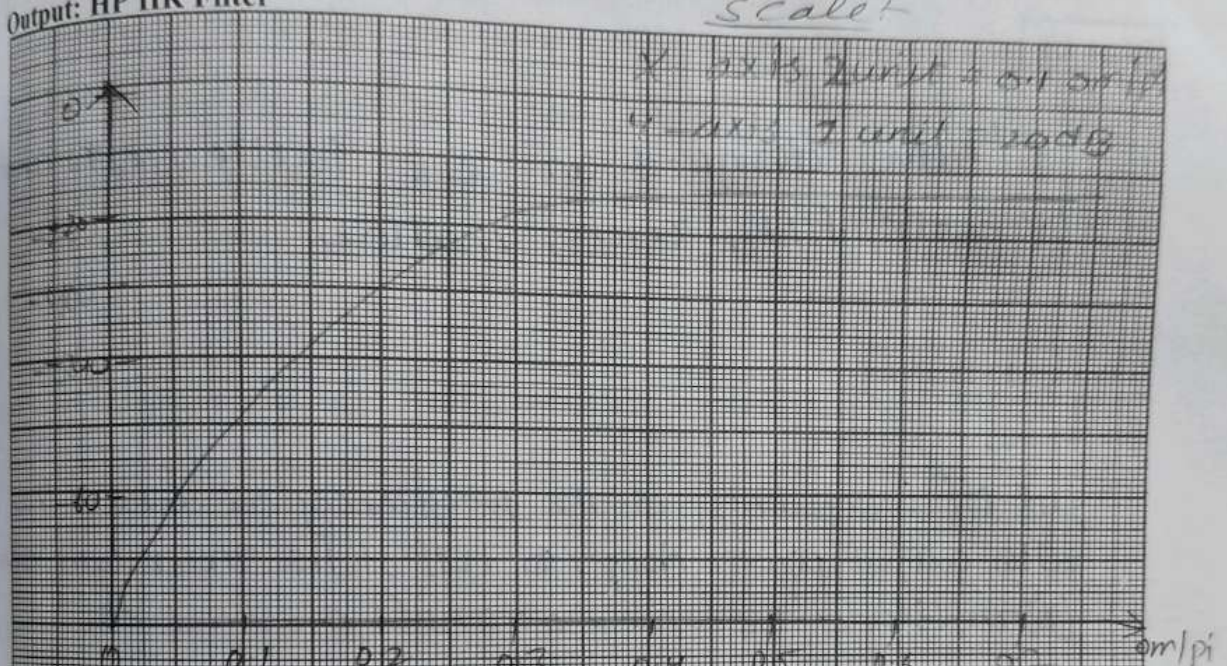
Output: LP IIR Filter

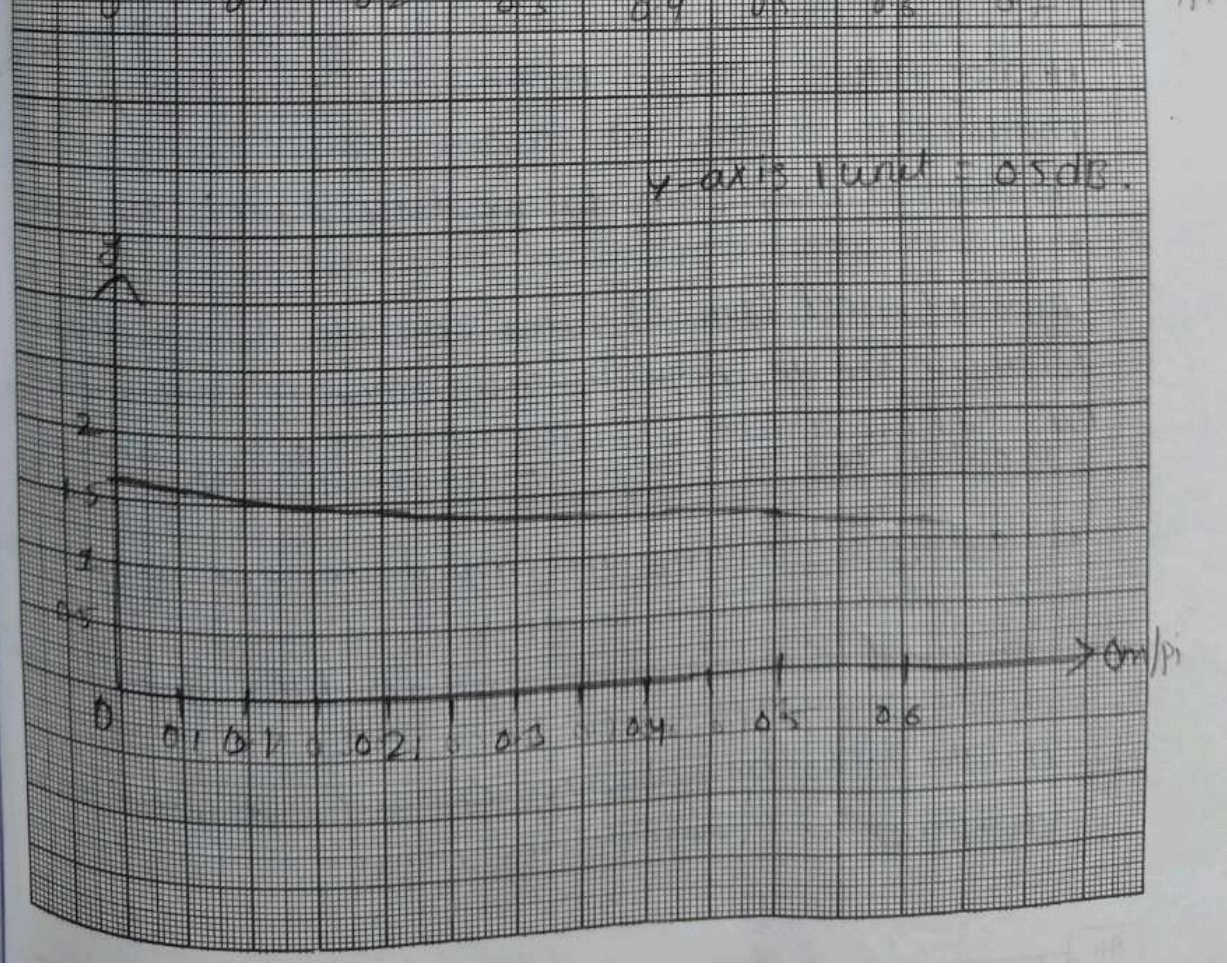
Scale:-



Output: HP IIR Filter

Scale:





BPF

Given specification

$\alpha_p = 2 \quad \alpha_s = 20 \quad f_{p1} = 1000 \quad f_{p2} = 800$

$f_{s1} = 2000 \quad f_{s2} = 2200$

$f_s = 8000$

$\omega = \frac{2\pi f}{f_s}$

$\omega_{p1} = \pi/4 \quad \omega_{p2} = \pi/5$

$\omega_{s1} = \pi/2 \quad \omega_{s2} = \frac{11\pi}{20}$

$\omega_p = \left[\frac{\pi}{4}, \frac{\pi}{5}\right]$

$\omega_s = \left[\frac{\pi}{2}, \frac{11\pi}{20}\right]$

$\omega_p = [0.7854, 0.6283]$

$\omega_s = [1.5708, 1.7279]$

$$N \geq \frac{\log\left(\frac{10^{0.1\alpha_s} - 1}{10^{0.1\alpha_p} - 1}\right)}{2\log\left(\frac{\omega_s}{\omega_p}\right)}$$

$N \geq 3.26$

$\boxed{N = 4}$

$$H(s) = \frac{(0.095)^4}{(s^2 + 0.09s + 0.134)^4} \qquad \boxed{H(s) = \frac{(Bs)^N}{(s^2 + Bs + \omega_0^2)^N}}$$

$$H(s) = \frac{0.00006565^4}{(s^2 + 0.9s + 0.134)^4}$$

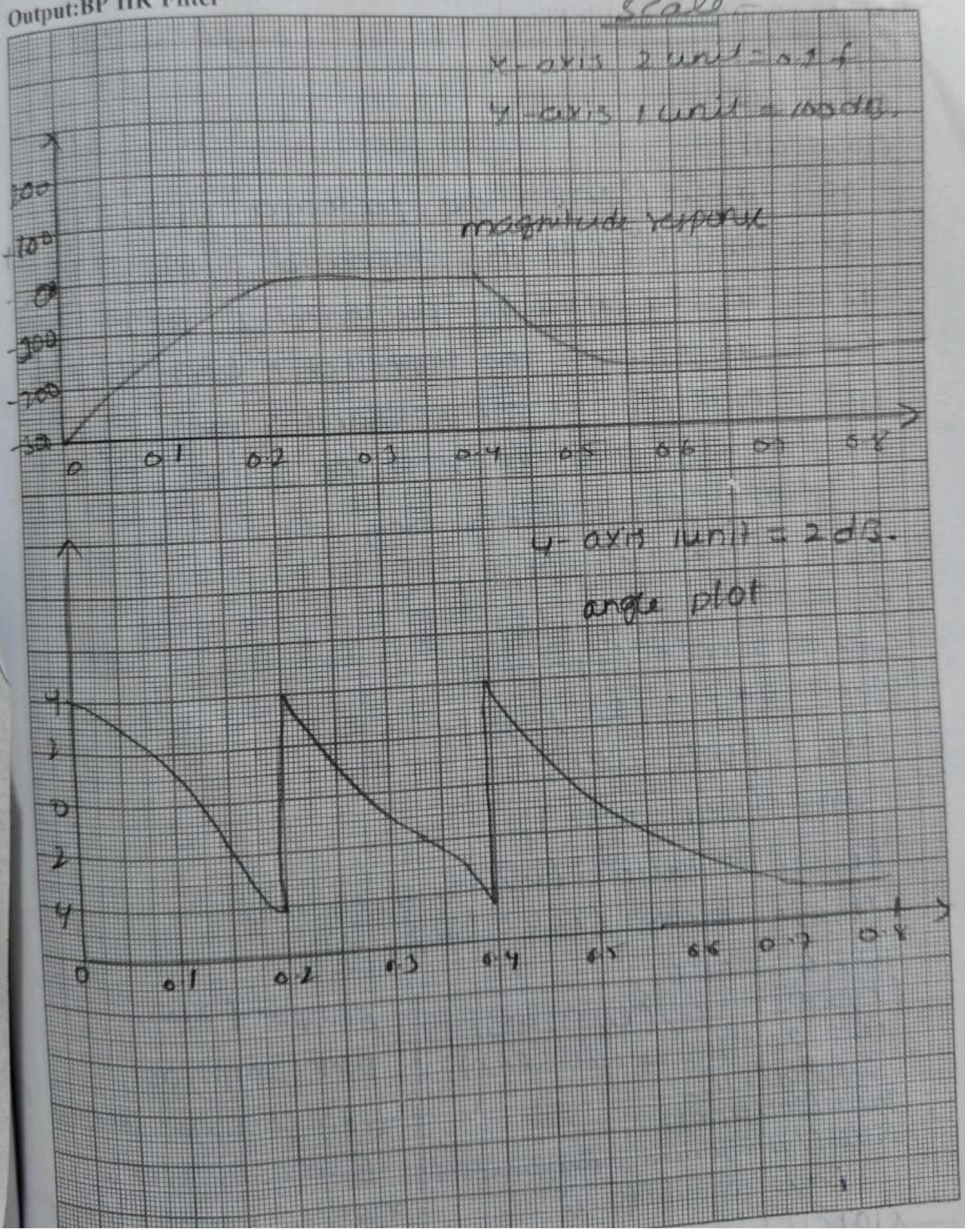
$$|H(j\omega)| = \frac{0.00006 56 (j\omega)^4}{((j\omega)^2 + 0.09(j\omega) + 0.134)^4}$$

$$|H(j\omega)| = \frac{0.00006 56 \omega^4}{(0.134 - \omega^2 + j0.09\omega)^4}$$

```
f = input('sampling freq :');
wp1 = 2 * pi * (fp1/f);
ws1 = 2 * pi * (fs1/f);
wp2 = 2 * pi * (fp2/f);
ws2 = 2 * pi * (fs2/f);
wp = [wp1 wp2];
ws = [ws1 ws2];
[n, wc] = buttord (wp, ws, rp, rs, 's');
[b, a] = butter (n, wc, 'bandpass', 's');
w = 0: 0.01: pi;
[h, om] = freqs (b, a, w);
m = 20* log10 ( abs(h));
subplot (211);
plot (om/pi, m);
title ('magnitude Response of Bandpass Filter
xlabel ('Normalized Frequency');
ylabel ('Gain in dB');
an = angle(h);
subplot (212);
plot(om/pi, an);
title ('phase Response of Bandpass Filter')
xlabel('Normalized Frequency');
ylabel('phase in radians');
```


Output:BP

Output:BP IIR Filter



MATLAB PROGRAM: BS IIR Filter

```
clc;
clear all;
close all;
rp = input('enter the passband ripple: ');
rs = input('enter the stopband ripple: ');
fp1 = input('enter the passband 1 freq: ');
fs1 = input('enter the stopband 1 freq: ');
fp2 = input('enter the passband 2 freq: ');
fs2 = input('enter the stopband 2 freq: ');
f = input('sampling freq: ');
wp1 = 2*pi*(fp1/f);
ws1 = 2*pi*(fs1/f);
wp2 = 2*pi*(fp2/f);
ws2 = 2*pi*(fs2/f);
wp = [wp1 wp2];
ws = [ws1 ws2];
[n,wc] = buttord(wp,ws,rp,rs,'s');
[b,a] = butter(n,wc,'stop','s');
w = 0:0.01:pi;
[h,om] = freqz(b,a,w);
m = 20*log10(abs(h));
subplot(211);
plot(om/pi,m);
title('magnitude Response of
    Band stop filter');
xlabel('Normalized frequency');
ylabel('Gain in dB');
an = angle(h);
subplot(212);
plot(om/pi,an);
title('phase response
    Band stop filter');
xlabel('Normalized freq
ylabel('phase in
    radians
```


Output: BS IIR

Theoritical ~~Calculations~~ Calculations

For Bandstop

$\omega_p = [0.7854, 0.6283]$

$\omega_s = [1.5708, 1.7279]$

$\Omega = \tan\left(\frac{\omega}{2}\right)$ $\quad \Omega_{p1} = 0.324$

$\Omega_{p2} = 1.727$

$\alpha_p = 2 \quad \alpha_s = 20$

$B = \Omega_{p2} - \Omega_{p1} = 1.403$

$\Omega_0 = \cancel{0.648}\sqrt{\Omega_{p1}\Omega_{p2}} = 0.748 \Rightarrow \Omega_0^2 = 0.559$

$$N \geq \frac{\log\left(\frac{10^{0.1\alpha_s}-1}{10^{0.1\alpha_p}-1}\right)}{2\log\left(\frac{\Omega_s}{\Omega_p}\right)}$$

$N \geq 5.8$

$\boxed{N=6}$

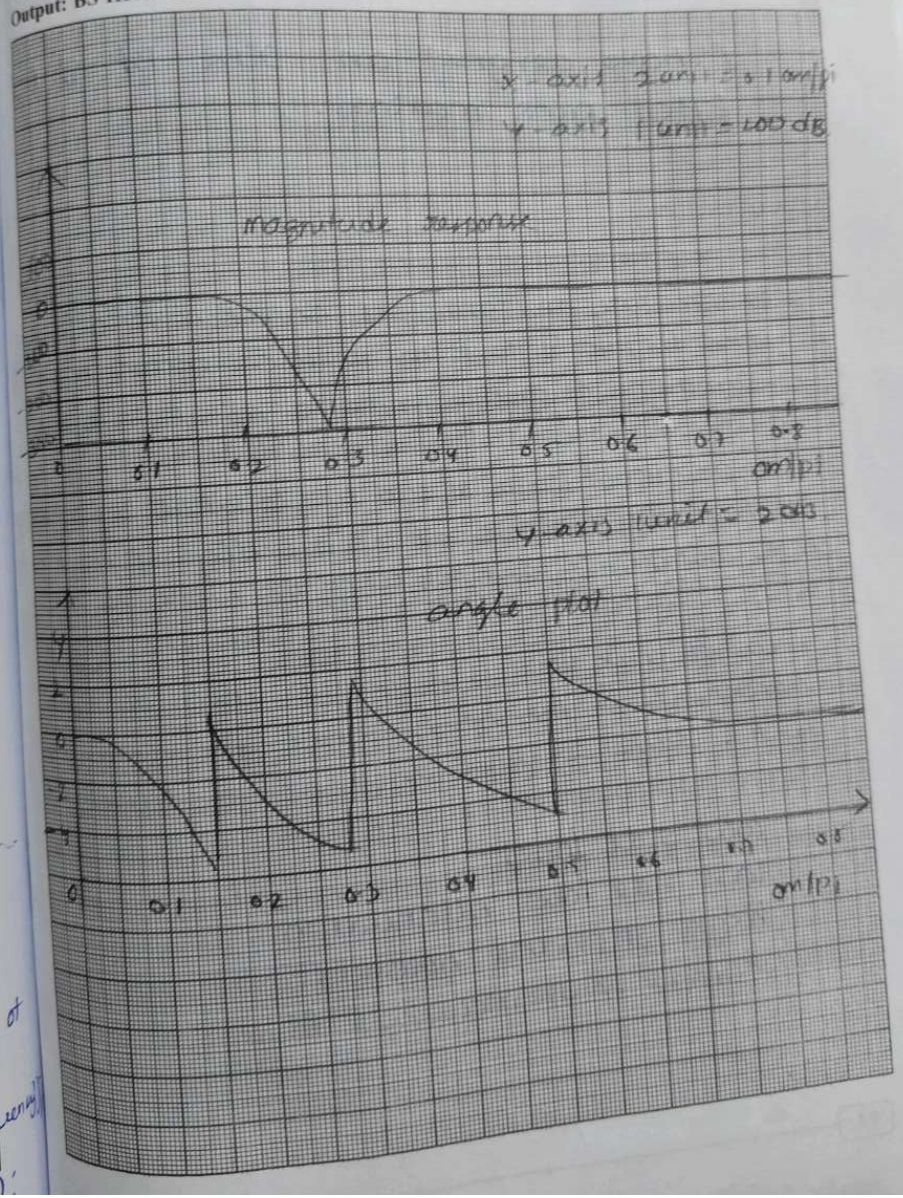
$$H(s) = \frac{(s^2+0.559)^6}{(s^2+1.403s+0.559)^6} \Rightarrow H(s) = \frac{(s^2+\Omega_0^2)^N}{(s^2+Bs+\Omega_0^2)^N}$$

$$H(j\omega) = \frac{(0.559-\omega^2)^6}{(0.559-\omega^2+j1.403\omega)^6}$$

$$|H(j\omega)| = \frac{|0.559-\omega^2|^6}{\left[\sqrt{(0.559-\omega^2)^2+(1.403\omega)^2}\right]^6}$$

$$\angle H(j\omega) = 6\tan^{-1}\left(\frac{1.403\omega}{0.559-\omega^2}\right)$$

Output: BS IIR Filter



EDURE

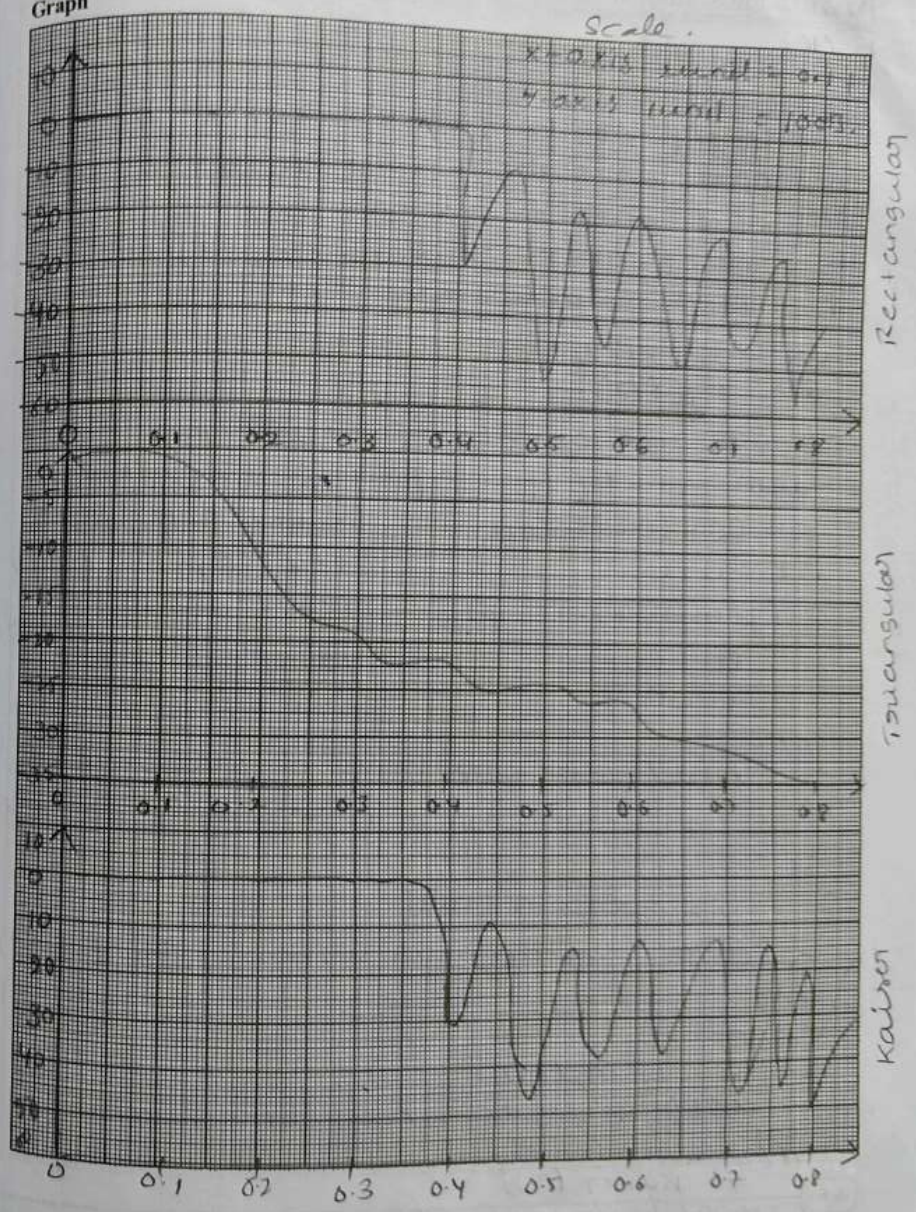
1. Open MATLAB
2. Open new M-file
3. Type the program
4. Save in current directory
5. Compile and Run the program
6. For the output see command window/ Figure window

MATLAB PROGRAM: LP FIR FILTER

```
clc;
clear all;
close all;
rp = input ('enter the passband ripple: ');
rs = input ('enter the stopband ripple: ');
fp = input ('enter the passband freq : ');
fs = input ('enter the stopband freq : ');
f = input ('sampling freq: ');
w1 = fp/(f/2);
w2 = fs/(f/2);
num = -20*log10(sqrt(rp*rs)) - 13;
dem = 14.6*(fs-fp)/f;
n = ceil(num/dem);
n1 = n+1;
if(rem(n,2)~=0)
    n1 = n;
    n = n-1;
end
c = input('enter your choice of window
function 1. Rectangular 2. Triangular
    3. Kaiser :');
if(c==1)
    y = rectwin(n1);
    disp('Rectangular window filter
response');
end
if(c==2)
    y = triang(n1);
    disp('Triangular window filter
response');
end
if(c==3)
    y = kaiser(n1);
    disp('Kaiser window filter response');
end
b = fir1(n,w1,y);
[h,o] = freqz(b,1,256);
m = 20*log10(abs(h));
subplot(211);
plot(o/pi,m);
title('Low pass filter');
xlabel('Normalized frequency');
ylabel('Gain in dB');
grid on;
```

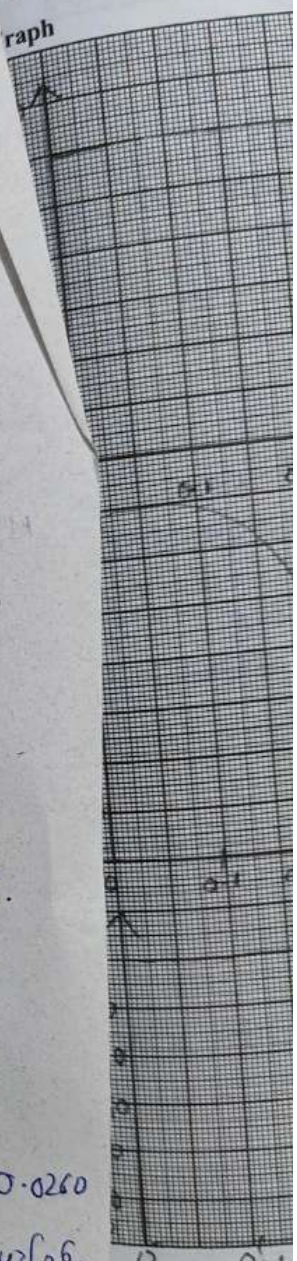

Output: LP FIR FILTER

Graph



utput: LP FIR FILTER

raph



Theoretical calculations (LP)

$\alpha p = 0.01$ $\alpha s = 0.001$ $fp = 1000$ $fs = 1500$ $f =$

$\omega p = 0.78539$ $\omega s = 1.17089$

$D\omega = 0.3927$ $\omega c = 0.98174$

$N = \frac{4\pi}{D\omega} = 31.99$ $\boxed{N = 31}$

$$hd(n) = \frac{1}{8\pi}\int_{-\omega c}^{\omega c}\left[\frac{e^{j0.9(n-\alpha)} - e^{-0.9j(n-\alpha)}}{j(n-\alpha)}\right]$$

$$= \frac{1}{(n-\alpha)\pi}\left[\sin 0.9(n-\alpha)\right]$$

$h(0) = 0.0170$

$h(1) = 0$

$h(2) = -0.0816$

$h(3) = -0.0260$

$h(4) = -0.0132$

$h(5) = 0.0131$

$h(6) = 0.0343$

$h(7) = 0.0315$

$h(8) = 0$

$h(9) = -0.0409$

$h(10) = -0.622$

$h(11) = -0.0352$

$h(12) = -0.0453$.

$h(13) = 0.1549$

$h(14) = 0.2493$ $h(15) = 0.3214$.

$WR = 1$.

$h(n) = hd(n)$

$$H(z) = z^{-\left(\frac{N-1}{2}\right)}\left[h(0) + \sum_{n=1}^{15} h(n)\left[z^{n} + z^{-n}\right]\right]$$

$$= z^{-15}\Big[0.0170 + 0\,(z^{-1}+z^{1}) - 0.0816(z^{-2}+z^{2}) - 0.0260 (z^{3}+z^{-3}) - 0.0132\,[z^{4}+z^{-4}] + 0.0131\,(z^{5}+z^{-5}) + 0.0343[z^{6}+ z^{-6}] + 0.0313\,(z^{7}+z^{-7}) + 0(z^{8}+z^{-8}) - 0.0409\,(z^{9}+z^{-9}) - 0.0632\,(z^{10}+z^{-10}) - 0.0354(z^{-11}+z^{11}) + 0.0453(z^{12}+z^{-12}) + 0.1549\,(z^{13}+z^{-13}) + 0.2493(z^{14}+z^{-14}) + 0.3214(z^{15}+z^{-15})$$

$a(0) = h(15) = 0.170$

$a(1) = 0$

$a(2) = 0.0372$

$a(3) = -0.052$.

$a(4) = 0.269$

$a(5) = 0.262$

$a(6) = 0.656$

$a(7) = 0.063$

$a(8) = 0$

$a(9) = -0.0818$

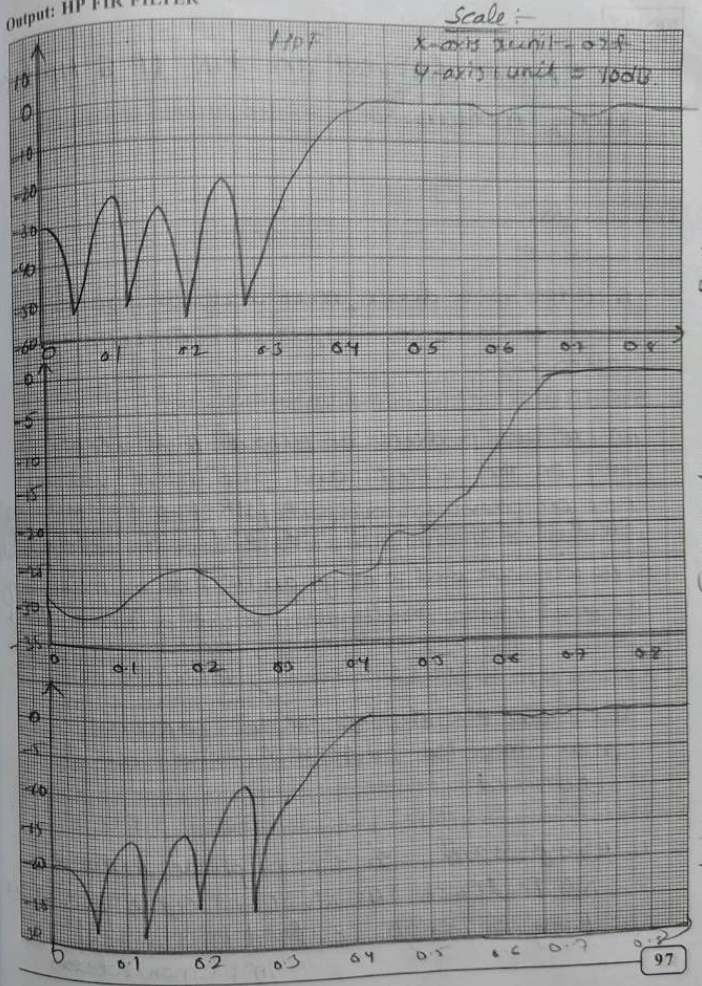
$a(10) = -0.1244$

$a(11) = -0.0704$

MATLAB PROGRAM:HP FIR FILTER

```
clc;
clear all;
close all;
rp = input('enter the passband ripple: ');
rs = input('enter the stopband ripple: ');
fp = input('enter the passband freq: ');
fs = input('enter the stopband freq: ');
f = input('sampling freq: ');
w1 = fp/(f/2);
w2 = fs/(f/2);
num = -20* log10(sqrt(rp * rs)) -13;
dem = 14.6 * (fp - fs)/f;
n = ceil(num/dem);
n1 = n+1;
if (rem(n,2)~= 0)
    n1=n;
    n=n-1;
end
c = input('enter your choice of window function
        1. Rectangular 2. Triangular 3. Kaiser:')
if(c==1)
    y = rectwin(n1);
    disp('Rectangular window filter response')
end
if(c==2)
    y = triang(n1);
    disp('triangular window filter response');
end
if(c==3);
    y = kaiser(n1);
    disp('kaiser window filter response');
b = fir1(n, w1,'high', y);
[h,o] = freq(b, 1, 256);
m = 20* log10(abs(h));
plot(o/pi, m);
title('High pass Filter');
xlabel('Normalized Frequency');
ylabel('Gain in db');
grid on;
```


Output: HP FIR FILTER



Theoretical (HP)

$\alpha p = 0.01$ $\alpha s = 0.001$ $f_{p1} = 2000$ $f_{s1} = 1000$

$f = 8000$

$\omega p1 = 1.57679$ $\omega s1 = 0.7853$

$\omega c = 1.178045$ $\Delta\omega = 0.78549$

$N = 15.998$ $\boxed{N=16}$

$$hd(n) = \left[\frac{\sin\pi(n-\alpha) - \sin\omega c(n-\alpha)}{\pi(n-\alpha)}\right]$$

$hd(0) = -0.0449$ $h(6) = -0.2837$

$hd(1) = -0.0165$ $h(7) = 0.6498$

$hd(2) = -0.0449$

$hd(3) = -0.0757$

$hd(4) = -0.0164$

$hd(5) = -0.1257$

$$H(7) = z^{-\left(\frac{N-1}{2}\right)}\left[h(0) + \sum_{n=1}^{7}\left(z^{n} + z^{-n}\right)\right]$$

$$= -0.0449z^{-7} - 0.0165\left(z^{-6} - z^{-8}\right) - 0.0497\left(z^{-5} + z^{-9}\right)$$
$$-0.0797\left(z^{-4} + z^{10}\right) - 0.0167\left(z^{-3} + z^{-11}\right) - 0.1257\left(z^{-2} + z^{12}\right)$$
$$-0.2837\left(z^{-1} + z^{-13}\right) + 0.649\left(z^{0} + z^{-14}\right)$$

$h(14) = 0.6498$

$h(1) = -0.2837$

$h(2) = -0.1287$

$h(3) = -0.0167$

$h(4) = -0.0757$

$h(5) = -0.0449$

$h(6) = -0.0165$

$h(7) = -0.0449$.

$a(0) = -0.0449$

$a(1) = -0.033$

$a(2) = -0.0698$

$a(3) = -0.1574$

$a(4) = -0.334$

$a(5) = -0.2574$

$a(6) = -0.5674$

$a(7) = 1.2996$.

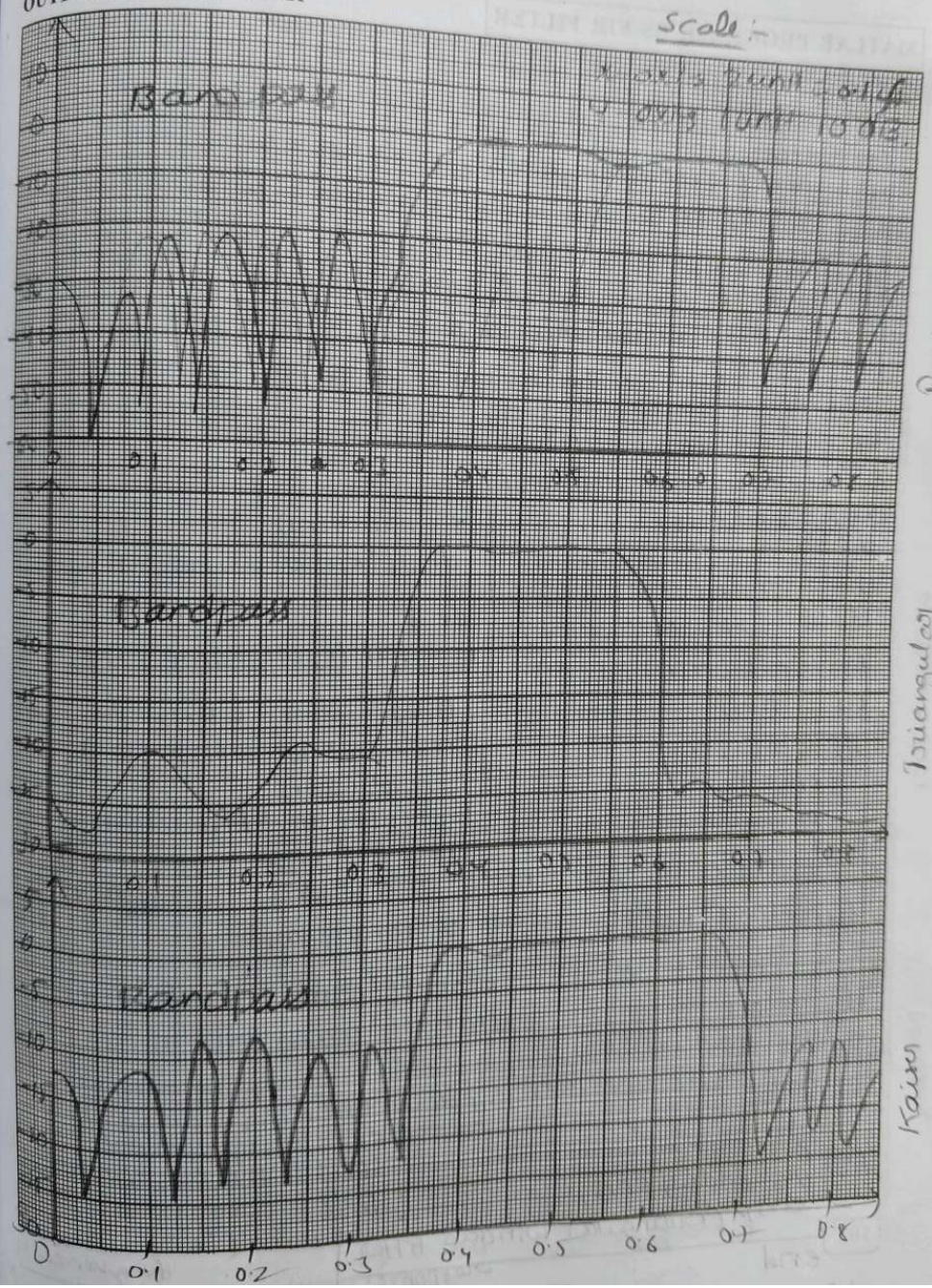
MATLAB PROGRAM

BP FIR Filter

```
clc;
clear all;
close all;
rp = input ('enter the passband ripple:');
rs = input ('enter the stopband ripple:');
fp1 = input ('enter the passband 1 freq:');
fs1 = input ('enter the stopband 1 freq:');
fp2 = input ('enter the passband 2 freq:');
fs2 = input ('enter the stopband 2 freq:');
f = input ('sampling freq:');
  N1 = [fp1 fp2]/(f/2);
  num = -20*log10(sqrt(rp*rs)) - 13;
  df1 = fp1 - fs1;
```

```
df2 = fs2 - fp2;
df = min (df1, df2);
dem = 14.6 * (df/f);
n = ceil (num/dem);
n1 = n+1;
if (rem(n,2) ~= 0)
    n1 = n;
    n = n-1;
end
c = input('enter your choice of window function
    1. Rectangular 2. Triangular 3. Kaiser:');
if (c==1)
    y = rectwin(n1);
    disp('Rectangular window filter response');
end
if(c==2)
    y = triang(n1);
    disp('Triangular window filter response');
end
if (c==3)
    y = Kaiser (n1);
    disp('Kaiser window filter response');
end
b = fir1 (n, w1, 'bandpass', y);
[h,o] = freq2 (b, 1, 256);
m = 20* log10 (abs(h));
subplot (211);
plot( o/pi, m);
title ('Band pass FIR filter');
xlabel('Normalized frequency');
ylabel('Gain in dB');
grid on;
```

OUTPUT:BP FIR FILTER



BS

$\alpha p = 0.01 \quad \alpha s = 0.001 \quad f_{p1} = 2000 \quad f_{s1} = 2500$

$f_{p2} = 4000 \quad f_{p1} = 4500.$

$\omega p_1 = 1.2566 \quad \omega s_1 = 0.9424 \quad \omega p_2 = 2.5132$

$\omega s_2 = 2.8074 \quad \omega c_1 = 1.0995 \quad \omega c_2 = 2.6704$

$D\omega = \omega p - \omega s = 0.3142$

$N = \frac{4\pi}{D\omega} = 41;$

$$hd(n) = \frac{1}{2\pi}\left[\frac{\sin\pi(n-\alpha) + \sin 1.0995(n-\alpha) \sin 2.75(n-\alpha)}{(n-\alpha)}\right]$$

$h(0) = h(40) = 0.0048 \quad h(11) = 0.0013$

$h(1) = h(39) = 0.0041 \quad h(12) = 0.0180$

$h(2) = 0.0281 \quad h(13) = 0.0304$

$h(3) = 0.0217 \quad h(14) = 0.0349$

$h(4) = 0.0173 \quad h(15) = 0.0306$

$h(5) = 0.0046 \quad h(16) = 0.0176$

$h(6) = 0.0105 \quad h(17) = 0.0014$

$h(7) = -0.0233 \quad h(18) = -0.0138$

$h(8) = -0.0294 \quad h(19) = -0.0240$

$h(9) = -0.0265 \quad h(20) = 0.522$

$h(10) = -0.0151$

$$H(z) = z^{-20}\left[h(0) + \sum_{n=1}^{20} h(n)\left(z^{n} + z^{-n}\right)\right]$$

$= 0.0048 z^{-20} + 0.0141\left[z^{-19} + z^{-21}\right] + 0.0216\left[z^{-18} + z^{-22}\right] + 0.0237\left[z^{-17} + z^{-23}\right] + 0.0173\left[z^{-16} + z^{-24}\right] + 0.0046\left[z^{-15} + z^{-25}\right] - 0.0105\left[z^{-14} + z^{-26}\right] - 0.0233\left[z^{-13} + z^{-27}\right] - 0.294\left(z^{-12} + z^{-28}\right) - 0.0265\left(z^{-11} + z^{-29}\right) - 0.0151\left(z^{-10} + z^{-30}\right) + 0.0013\left[z^{-9} + z^{-31}\right] + 0.0180\left[z^{-8} + z^{-12}\right] +$

$$0.0304\left[z^{-7}+z^{-13}\right]+0.0349\left[z^{-6}+z^{-14}\right]+\left[z^{-5}+z^{-15}\right]+0.0176\left[z^{-4}+z^{-36}\right]+0.0014\left[\right.$$

$$-0.0138\left[z^{-2}+z^{-38}\right]-0.0240\left[z^{-1}+z^{-39}\right]+0.522\left[z^{0}+\right.$$

```
dt = n
dt = 14
n = ceil (
n1 = n+1
if (rem(n
n1
```

$a(0) = 0.0048$

$a(1) = 0.0282$

$a(2) = 0.0218$

$a(3) = 0.0474$

$a(4) = 0.0346$

$a(5) = 0.009$

$a(6) = 0.021$

$a(7) = 0.0466$

$a(8) = -0.0588$

$a(9) = -0.053$

$a(10) = -0.00302$

$a(11) = 0.002$

$a(12) = 0.036$

$a(13) = 0.0608$

$a(14) = 0.0698$

$a(15) = 0.0602$

$a(16) = 0.0352$

$a(17) = 0.002$

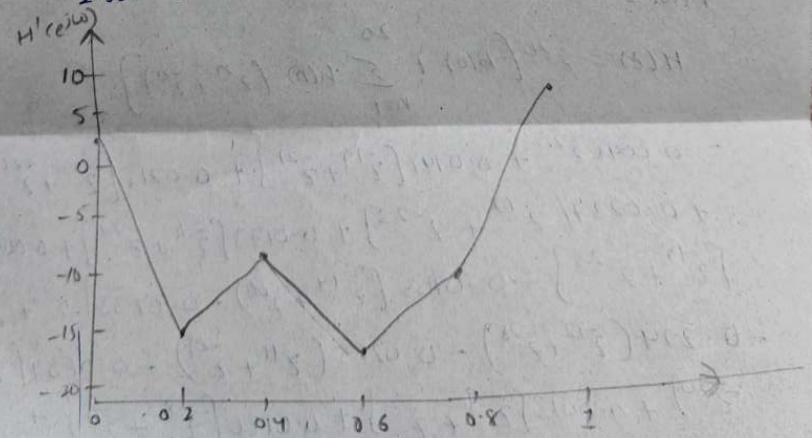
$a(18) = 0.0276$

$a(19) = -0.048$

$a(20) = 1.044$

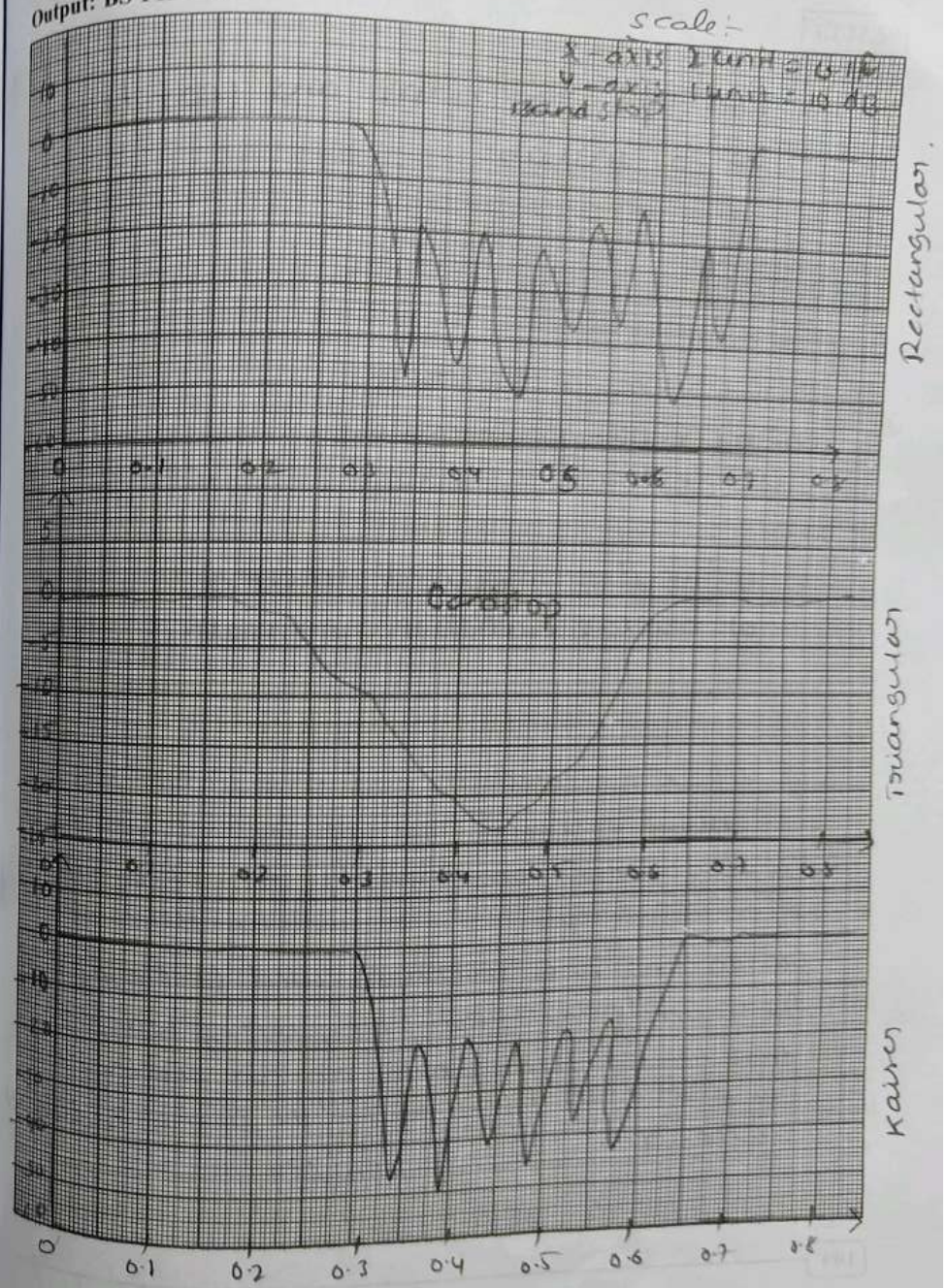
$$H'(e^{j\omega}) = 0.0048 + 0.0282\cos\omega + 0.0218\cos 2\omega + 0.0474\cos 3\omega + 0.0346\cos 4\omega + 0.009\cos 5\omega + 0.021\cos 6\omega + 0.0466\,7\omega - 0.0588\cos 8\omega - 0.053\cos 9\omega - 0.0302\cos 10\omega + 0.002\cos 11\omega + 0.36\cos 12\omega + 0.068\cos 13\omega + 0.069\cos 14\omega + 0.6\cos 15\omega + 0.0352\cos 16\omega + 0.002\cos 17\omega - 0.021\cos 18\omega - 0.48\cos 19\omega + 1.044\cos 20\omega.$$

ω	0	0.2	0.4	0.6	0.8	1
	2.52	-16.92	-9	-18.57	-11.42	6.72



MATLAB PROGRAM: BS FIR FILTER

```
clc;
clear all;
close all;
rp = input('enter the passband ripple: ');
rs = input('enter the stopband ripple: ');
fp1 = input('Enter the passband 1 freq: ');
fs1 = input('enter the stopband 1 freq: ');
fp2 = input('enter the passband 2 freq: ');
fs2 = input('enter the stopband 2 freq: ');
f = input('sampling freq: ');
w1 = [fs1 fs2]/(f/2);
num = -20*log10(sqrt(rp*rs)) -13;
df1 = abs(fp1 - fs1);
df2 = abs(fs2 - fp2);
df = min(df1, df2);
dem = 14.6*(df/f);
n = ceil(num/dem);
n1 = n+1;
if rem(n,2)~=0
    n1 = n;
    n = n-1;
end
c = input('enter your choice of window function 1. Rectangular 2. Triangular
  3. Kaiser :');
if c==1
    y = rectwin(n1);
    disp('Rectangular window filter response');
end
if c==2
    y = triang(n1);
    disp('triangular window filter response');
end
if c==3
    y = kaiser(n1);
    disp('kaiser window filter response');
end
b = fir1(n, w1, 'stop', y);
[h,o] = freqz(b,1,512);
m = 20*log10(abs(h));
subplot(211);
plot(o/pi, m, 'Linewidth', 1.5);
title('Band stop FIR filter- magnitude response');
xlabel('normalized freq (1 time)');
ylabel('Gain (dB)');
grid on;
% phase response
an = angle(h)
subplot(212);
plot(o/pi, an, 'Linewidth', 1.5);
title('Band stop fir filter');
xlabel('normalized frequency');
ylabel('phase (radian)');
grid on;
```


Output: BS FIR Filter